

1. PROBLEM STATEMENT

Forests are an essential part of the environment; they house habitats for countless plants and animals, provide natural resources and recreational areas, and absorb enormous amounts of carbon dioxide. However, forests are increasingly afflicted by drought, pests, and disease. Current methods for determining a tree's health are time consuming and are often commissioned only after visible damage is observed, or they harm the tree during testing. A solution is needed to address this problem by providing a low cost, reliable, and nondestructive method to continuously monitor tree health throughout forested areas.

1.1. Need Statement

According to the U.S Department of Agriculture nearly 11% of forests in the United States are exposed to drought, expected to rise to 38% by mid-century, while 11-16% of tree species are being driven to extinction by insects and disease [1], [2]. Although methods of monitoring tree health already exist, they require foresters to manually assess trees by estimating water content in the leaves or cutting down trees to measure biomass. These methods are often time-consuming with each plot usually taking one day to complete, depending on the plot size [3]. Additionally, these methods require visible indicators of tree decay leading to mean tree health initiatives being often too late for any individual tree or potentially an entire surrounding population. Many problems afflicting forested populations, from vascular wilts to root rot, are not easily or economically detectable by other methods of observation. A portable, reliable, and nondestructive device for monitoring tree health is needed to allow forest health to be assessed more efficiently.

1.2. Objective Statement

The objective of this project is to design and build a robot that monitors the health of trees and forests. The robot measures signal attenuation from GPS satellites to estimate water content in branches and leaves. The Forest Buddy is guided by waypoints, set by the user walking through the forest, but can be operated through a remote control if necessary. As the Forest Buddy collects data, it sends this data to a database where it can be compared with control data from a base station which is in an open field. Researchers or land agencies can then use this data to estimate the health of forests.

1.3. Background and Related Work

Currently, forest health is assessed using manual inspection, lab testing, drone networks, and fixed sensor networks. Manual inspection of forests is impractical for large areas, and usually only detects illness in the symptomatic stage, which is not preferable [4], [5]. While lab testing is fast and effective, it requires the tree to be cut. Advanced technological methods detect problems earlier; however, such methods have their own problems [4]. Remote sensing techniques such as drone and satellite imaging can struggle with environmental conditions and coverage limitations that make consistent, large-scale monitoring difficult [4], [5]. Fixed sensor networks can continuously monitor tree health, but require installing a device on every tree, which is prohibitively expensive to scale to large areas [4]. Furthermore, it must be mentioned that Forest Buddy has the same core concepts as a previous project called Leaf Link, which is limited by a pulley system and cannot cover many trees. On the other hand, Forest Buddy will be effective at detecting sickness in a time efficient, inexpensive, non-destructive, and scalable way with its advantage of mobility to get more data and save more trees.

References

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